43.2: In adaptive immunity, receptors provide pathogen-specific recognition

Vertebrates are unique in having adaptive as well as innate immunity. In adaptive immunity, molecular recognition relies on a vast arsenal of receptors, each of which recognizes a feature typically found only on a particular part of a particular molecule in a particular pathogen. As a result, recognition and response in adaptive immunity occur with remarkable specificity.

The adaptive immune response, also known as the acquired immune response, is activated after the innate immune response and develops more slowly. Unlike innate immunity, the adaptive response is enhanced by previous exposure to the infecting pathogen. **Figure 43.7** highlights these and other fundamental similarities and differences between adaptive and innate immunity.

Figure 43.7

Overview of animal immunity.

An overview of animal immunity.

Innate immunity offers a primary defense in all animals and sets the stage for adaptive immunity in vertebrates.

Innate immunity (all animals):

- recognition of traits shared by broad ranges of pathogens, using a small set of receptors
- rapid response

Under the umbrella of innate immunity are barrier defenses and internal defenses. Pathogens (such as bacteria, fungi, and viruses) are included here.

Barrier defenses:

- Skin
- mucous membranes
- secretions

Internal defenses:

- phagocytic cells
- natural killer cells

- antimicrobial proteins
- inflammatory response

Adaptive immunity (vertebrates only):

- recognition of traits specific to particular pathogens using a vast array of receptors
- slower response

Under the umbrella of adaptive immunity are humoral response and cell-mediated response, the former of which is possessed by pathogens.

Humoral response:

- antibodies defend against infection in body fluids

Cell-mediated response:

- cytotoxic cells defend against infection in body cells.

Adaptive immunity relies on T cells and B cells, which are types of white blood cells called **lymphocytes(Figure 43.8)**. Like all blood cells, lymphocytes originate from stem cells in the bone marrow. Some migrate from the bone marrow to the **thymus**, an organ in the thoracic cavity above the heart (see Figure 43.6). These lymphocytes mature into **T cells**. Lymphocytes that remain and mature in the bone marrow develop as **B cells**. (Lymphocytes of a third type also mature in bone marrow to become the natural killer cells active in innate immunity.)

Figure 43.8

B and T lymphocytes.

Mature B cells and T cells. B cells have Y-shaped antigen receptors on the surface and T cells have antigen receptors with two polypeptide chains on the surface

Antigens as the Trigger for Adaptive Immunity

Any substance that elicits a B or T cell response is an **antigen**. In adaptive immunity, recognition occurs when a B cell or T cell binds to an antigen, such as a bacterial or viral protein, via a protein called an **antigen receptor**. Each antigen receptor binds to just one part of one molecule from a particular pathogen, such as a species of bacteria or strain of virus.

The cells of the immune system produce millions of different antigen receptors. A given lymphocyte, however, produces just one variety; all of the antigen receptors made by a single B or T cell are identical. Infection by a virus, bacterium, or other pathogen triggers activation of B and T cells with antigen receptors specific for parts of that pathogen. Although drawings of B and T cells typically include just a few antigen receptors, a single B or T cell actually has about 100,000 antigen receptors on its surface.

Antigens are usually foreign and are typically large molecules, either proteins or polysaccharides. Many antigens protrude from the surface of foreign cells or viruses. Other antigens, such as toxins secreted by bacteria, are released into the extracellular fluid.

The small, accessible portion of an antigen that binds to an antigen receptor is called an **epitope**. An example is a group of amino acids in a particular protein. A single antigen usually has several epitopes, each binding a receptor with a different specificity. Because all antigen receptors produced by a single B cell or T cell are identical, they bind to the same epitope. Each B or T cell thus displays *specificity* for a particular epitope, enabling it to respond to any pathogen that produces molecules containing that epitope.

The antigen receptors of B cells and T cells have similar components, but they encounter antigens in different ways. We'll consider the two processes in turn.

Antigen Recognition by B Cells and Antibodies

Each B cell antigen receptor is a Y-shaped protein consisting of four polypeptide chains: two identical **heavy chains** and two identical **light chains(Figure 43.9)**. Disulfide bridges link the chains together.

Figure 43.9

The structure of a B cell antigen receptor.

Y-shaped B-cell antigen receptor on surface of a B cell. Variable regions of the heavy and light chains at the tip of the Y make up the antigen binding sites.

Y-shaped B-cell antigen receptor on the surface of a B cell.

The Y is composed of two identical heavy chains, with each forming half of the tail or stem of the Y and one half of the two arms that compose the top (V shaped part) of the Y receptor. The transmembrane region on the tip of the tail of the Y spans the plasma membrane of the B cell. The tail of each heavy chain is composed of long constant c. regions joined by disulfide bridges. Near the V shaped tips of each heavy chain are short variable (V) regions. Parallel to each variable region in the heavy chain are the shorter light chains. Each light chain consists of a constant c. region near the base joined to the heavy chain constant regions with disulfide bridges, with variable regions at the tips alongside the variable regions in the heavy chains. Antigen binding sites are the variable (V) tips of both the heavy and light chains at the top of the Y on each side.

Each light chain or heavy chain has a *constant*(C)*region*, where amino acid sequences vary little among the receptors on different B cells. The constant region of heavy chains contains a transmembrane region, which anchors the receptor in the cell's plasma membrane. As shown in Figure 43.9, each light or heavy chain also has a *variable*(V)*region*, so named because its amino acid sequence varies extensively from one B cell to another. Together, parts of a heavy-chain V region and a light-chain V region form an asymmetric binding site for an antigen. Therefore, each B cell antigen receptor has two identical antigen-binding sites.

Binding of a B cell antigen receptor to an antigen is an early step in B cell activation, leading eventually to formation of cells that secrete a soluble form of the receptor **(Figure 43.10a)**. This secreted protein is called an **antibody**, also known as an **immunoglobulin (Ig)**. Antibodies have the same Y-shaped structure as B cell antigen receptors but lack a membrane anchor. As you'll see later, antibodies provide a direct defense against pathogens in body fluids.

Figure 43.10

Antigen recognition by B cells and antibodies.

Antigen recognition by B cells and antibodies.

Parts are as follows.

- Part a. B cell antigen receptors and antibodies. An antigen receptor of a B cell binds to an epitope, a particular part of an antigen. Diagram depicts epitope regions on the surface of antigens as having complementary shapes to the antigen binding sites of B cell antigen receptors, so that the pathogen epitopes and the antigen receptor binding sites fit together. Following binding, the B cells give rise to cells that secrete a soluble form of the antigen receptor. This soluble receptor, called an antibody, is specific for the same epitope as the original B cell. Diagram depicts these antibodies as having the same shape as the antigen receptor on B cells, with the same shape antigen binding sites, except the antibodies are not attached to B cells.
- Part b. Antigen receptor specificity. Different antibodies can recognize free antigens as well as antigens on a pathogen's surface. In this diagram, antibody A with a triangular antigen receptor binds to a complementary epitope on the surface of a pathogen; antibody B with a rounded antigen receptor binds to a different complementary epitope region on the same pathogen. Antibody C with square antigen receptors binds complementary epitope regions on two free antigens with the antigen receptors on each of its arms.

MAKE CONNECTIONS *The interactions depicted here involve a highly specific binding between antigen and receptor (see also*Figure 5.17*)*. How is this type of interaction similar to an enzyme-substrate interaction (see**Figure 8.15**)?*

*For suggested answer, see*Appendix A.**

The antigen-binding site of a membrane-bound receptor or antibody has a unique shape that provides a lock-and-key fit for a particular epitope. This stable interaction involves many noncovalent bonds between an epitope and the surface of the binding site. Differences in the amino acid sequences of variable regions provide the variation in binding surfaces that enables binding to be highly specific.

B cell antigen receptors and antibodies bind to intact antigens in the blood and lymph. As illustrated in **Figure 43.10b** for antibodies, they can bind to antigens on the surface of pathogens or free in body fluids.

Antigen Recognition by T Cells

For a T cell, the antigen receptor consists of two different polypeptide chains, an α*chain* and a β*chain*, linked by a disulfide bridge **(Figure 43.11)**. Near the base of the T cell antigen receptor (often called simply a T cell receptor) is a transmembrane region that anchors the molecule in the cell's plasma membrane. At the outer tip of the molecule, the variable (V) regions of the α and β chains together form a single antigen-binding site. The remainder of the molecule is made up of the constant (C) regions.

Figure 43.11

The structure of a T cell antigen receptor.

Antigen receptors consist of alpha and beta chains, linked by a disulfide bridge. Constant regions c. span the plasma membrane. Antigen binding sites are in the variable (V) regions.

Antigen receptors on the surface of a T cell consist of two side-by-side chains, the alpha chain and the beta chain, linked by a disulfide bridge. The transmembrane regions of each chain span the T cell plasma membrane. Long constant c regions are present at the bottom of both chains, with shorter variable (V) regions at the tips of the chains. At the very tips of the V regions are the antigen binding sites.

Whereas the antigen receptors of B cells bind to epitopes of *intact* antigens protruding from pathogens or circulating free in body fluids, antigen receptors of T cells bind only to fragments of antigens that are displayed, or presented, on the surface of host cells. The host protein that displays the antigen fragment on the cell surface is called a **major histocompatibility complex molecule(MHC).** By displaying antigen fragments, MHC molecules are essential for antigen recognition by T cells.

The display of protein antigens occurs when a pathogen infects a cell of the animal host or when an immune cell engulfs pathogen proteins or a whole pathogen **(Figure 43.12)**. Inside the animal cell, enzymes cleave each antigen into antigen fragments, which are small peptides. Each antigen fragment binds to an MHC molecule, which transports the bound peptide to the cell surface. The result is **antigen presentation**, the display of the antigen fragment in an exposed groove of the MHC protein.

Figure 43.12

Antigen recognition by T cells.

Antigen recognition by a T cell.

Inside the host cell, an antigen fragment from a pathogen binds to an MHC molecule and is brought up to the cell surface, where it is displayed. The combination of MHC molecule and antigen fragment is recognized by a T cell.

A host cell takes in a pathogen and, inside the host cell, enzymes cleave the antigen into fragments. An antigen fragment binds to an MHC molecule and is brought up to the cell surface, where it is displayed. The combination of MHC molecule and antigen fragment is recognized by a T cell when a T cell antigen receptor binds to the antigen fragment displayed by an MHC molecule.

Figure 43.13 shows a close-up view of antigen presentation. In effect, antigen presentations advertises the fact that a host cell contains a foreign substance. If the cell displaying an antigen fragment encounters a T cell with the right specificity, the antigen receptor on the T cell can bind to both the antigen fragment and the MHC molecule. This interaction of an MHC molecule, an antigen fragment, and an antigen receptor triggers an adaptive immune response, as we'll explore in Concept 43.3.

Figure 43.13

A closer look at antigen presentation.

A closer look at antigen presentation.

As shown in this ribbon model, the top of the MHC molecule cradles an antigen fragment, like a bun holding a hot dog. An MHC molecule can display many different antigen fragments, but the antigen receptor of a T cell is specific for a single antigen fragment.

The top of the MHC molecule cradles an antigen fragment, like a bun holding a hot dog. An MHC molecule can display many different antigen fragments, but the antigen receptor of a T cell is specific for a single antigen fragment.

B Cell and T Cell Development

Now that you know how B cells and T cells recognize antigens, let's consider four major characteristics of adaptive immunity. First, the immense repertoire of lymphocytes and receptors enables detection of antigens and pathogens never before encountered. Second, adaptive immunity normally has self-tolerance, the lack of reactivity against an animal's own molecules and cells. Third, cell proliferation triggered by

activation greatly increases the number of B and T cells specific for an antigen. Fourth, there is a stronger and more rapid response to an antigen encountered previously, due to a feature known as *immunological memory*, which we'll explore later in the chapter.

Receptor diversity and self-tolerance arise as a lymphocyte matures. Cell proliferation and the formation of immunological memory occur later, after a mature lymphocyte encounters and binds to a specific antigen. We'll consider these four characteristics in the order in which they develop.

The Basis of B Cell and T Cell Diversity

Each person makes more than 1million different B cell antigen receptors and 10 million different T cell antigen receptors. Yet there are only about 20,000 protein-coding genes in the human genome. How, then, do we generate so many different antigen receptors? The answer lies in combinations. Think of selecting a cell phone model that comes in three sizes and six colors. There are 18(3×6) combinations to consider. Similarly, by combining variable elements, the immune system assembles millions of different receptors from a very small collection of parts.

To understand the origin of receptor diversity, let's consider an immunoglobulin (Ig) gene that encodes the light chain of both membrane-bound B cell antigen receptors and secreted antibodies (immunoglobulins). Although we'll analyze only a single Ig light-chain gene, all B and T cell antigen receptor genes undergo very similar transformations.

The capacity to generate diversity is built into the structure of Ig genes. A receptor light chain is encoded by three gene segments: a variable (V) segment, a joining (J) segment, and a constant (C) segment. The V and J segments together encode the variable region of the receptor chain, while the C segment encodes the constant region. The light-chain gene contains a single C segment, 40 different V segments, and 5 different J segments. The alternative copies of the V and J segments are arrayed along the gene in a series **(Figure 43.14)**. Because a functional gene is built from one copy of each type of segment, the pieces can be combined in 200 different ways (40V×5J×1C). The number of different heavy-chain combinations is even greater, resulting in even more diversity.

Figure 43.14

Immunoglobulin (antibody) gene rearrangement.

Recombination of undifferentiated B cell DNA with V, J and C regions produces functional gene that is transcribed and translated into new light chain polypeptides on B cell antigen receptor.

The DNA of an undifferentiated B cell has regions (from left to right) V 37, V 38, V 39, and V 40 separated by longer introns, and regions J 1, J 2, J 3, J 4, and J 5 separated by shorter introns. An intron also separates J 5 from a C region at the far right of the DNA.

- Recombination deletes DNA between randomly selected V and J segments (in this example, DNA between V 39 and J 5 is randomly deleted). The result is DNA of a differentiated B cell, which now consists of (from left to right): V 37 and V 38 separated by an intron, followed by another intron separating V 38 from the functional gene, which consists of V 39 adjacent to J 5, followed by an intron and then a C region.
- Transcription of the permanently rearranged, functional gene from step 1 results in a pre-mRNA molecule with the regions (from left to right): V 39, J 5, intron, C
- RNA processing (removal of intron; addition of cap and poly-A tail) leads to mRNA molecule (left to right): Cap, V 39, J 5, C, poly-A tail.
- After translation, the light-chain polypeptide is produced, consisting of a variable region and a constant region. The light chain polypeptides are parallel to the top (V shaped part) of the antigen-receptors on the surface of B cells, with the variable region of the light chain lined up with the variable region at the tip of the heavy chain and the constant region of the light chain joined by a disulfide bridge to the top of the constant region of the heavy chain (refer to Figure 43.9).

The joining of randomly selected V and J gene segments (V39 and J5 in the example shown) results in a functional gene that encodes the light-chain polypeptide of a B cell antigen receptor. Transcription, splicing, and translation result in a light chain that combines with a polypeptide produced from an independently rearranged heavy-chain gene to form a functional receptor. Mature B cells (and T cells) are exceptions to the generalization that all diploid cells in the body have exactly the same DNA.

MAKE CONNECTIONS *Both alternative splicing and joining of* V *and* J *segments by recombination generate diverse gene products from a limited set of gene segments (see*Figure 18.14*)*. How do these processes differ?*

*For suggested answer, see*Appendix A*.*

Antigen Receptor Gene Rearrangement

Assembling a functional Ig gene requires rearranging the DNA. Early in B cell development, an enzyme complex called *recombinase* links one light-chain V gene segment to one J gene segment. This recombination event eliminates the long stretch of DNA between the segments, forming a single exon that is part V and part J.

Recombinase acts randomly, linking any one of the 40V gene segments to any one of the 5J gene segments. Heavy-chain genes undergo a similar rearrangement. In any given cell, however, only one allele of a light-chain gene and one allele of a heavy-chain gene are rearranged. Furthermore, the rearrangements are permanent and are passed on to the daughter cells when the lymphocyte divides.

After both a light-chain and a heavy-chain gene have been rearranged, antigen receptors can be synthesized. The rearranged genes are transcribed, and the transcripts are processed for translation. Following translation, the light chain and heavy chain assemble together, forming an antigen receptor (see Figure 43.14). Each pair of randomly rearranged heavy and light chains results in a different antigen-binding site. For the total population of B cells in a human body, the number of such combinations has been calculated as 3.5×106. Furthermore, mutations introduced during VJ recombination add additional variation, making the number of antigen-binding specificities even greater.

Origin of Self-Tolerance

In adaptive immunity, how does the body distinguish self from nonself? Because antigen receptor genes are randomly rearranged, some immature lymphocytes produce receptors specific for epitopes on the organism's own molecules. If these self-reactive lymphocytes were not eliminated or inactivated, the immune system could not distinguish self from nonself and would attack body proteins, cells, and tissues. Instead, as lymphocytes mature in the bone marrow or thymus, their antigen receptors are tested for self-reactivity. Some B and T cells with receptors specific for the body's own molecules are destroyed by *apoptosis*, which is a programmed cell death (see Concept 11.5). The remaining self-reactive lymphocytes are typically rendered nonfunctional, leaving only those lymphocytes that react to foreign molecules. Since the body normally lacks mature lymphocytes that can react against its own components, the immune system is said to exhibit *self-tolerance*.

Proliferation of B Cells and T Cells

Despite the enormous variety of antigen receptors, only a tiny fraction are specific for a given epitope. How then does an effective adaptive response develop? To begin with, an antigen is presented to a steady stream of lymphocytes in the lymph nodes (see Figure 43.6) until a match is made. A successful match between an antigen receptor and an epitope initiates events that activate the lymphocyte bearing the receptor.

Once activated, a B cell or T cell undergoes multiple cell divisions. For each activated cell, the result of this proliferation is a clone, a population of cells that are identical to the original cell. Some cells from this clone become **effector cells**, mostly short-lived cells that take effect immediately against the antigen and any pathogens producing that antigen.

For B cells, the effector forms are *plasma cells***(Figure 43.15)**, which secrete antibodies. For T cells, the effector forms are helper T cells and cytotoxic T cells, whose roles we'll explore in Concept 43.3. The remaining cells in the clone become **memory cells**, long-lived cells that can give rise to effector cells if the same antigen is encountered later in the animal's life.

Figure 43.15

A plasma cell.

A plasma cell has several distinct regions. It is on a 2 micrometer scale.

Figure 43.16 summarizes the proliferation of lymphocyte into a clone of cells occurs in response to binding to an antigen, using B cells as an example. The process is called **clonal selection** because an encounter with an antigen *selects* which lymphocyte will divide to produce a *clonal* population of thousands of cells specific for a particular epitope.

Figure 43.16

Clonal selection of B cells.

Clonal selection of B cells.

The process is as follows.

- Antigens bind to the antigen receptors of only one of the three B cells diagrammed, as each B cell differs in specificity based on the shape of its antigen receptors.
- The selected B cell (only the one that bound the antigen) proliferates, forming a clone of identical cells bearing receptors for the antigen.
- Some daughter cells develop into long-lived memory cells that can respond rapidly upon subsequent exposure to the same antigen, and other daughter cells develop into short-lived plasma cells that secrete antibodies specific for the antigen encountered.

VISUAL SKILLS *For the purpose of illustration, this figure shows only a few of each type of cell or molecule. Based on what you have read in*Concept 43.2*, provide estimates of the number of different B cells and number of antigen receptors on each B cell.*

*For suggested answer, see*Appendix A.*

Animation

Clonal Selection

image

Play

image

Play

Immunological Memory

Immunological memory is responsible for the long-term protection that a prior infection provides against many diseases, such as chicken pox. This type of protection was noted almost 2,400years ago by the Greek historian Thucydides. He observed that individuals who had recovered from the plague could safely care for those who were sick or dying, “for the same man was never attacked twice—never at least fatally.”

Prior exposure to an antigen alters the speed, strength, and duration of the immune response. The effector cells formed by clones of lymphocytes after an initial exposure to an antigen produce a **primary immune response**. The primary response peaks about 10–17 days after the initial exposure. If the same antigen is encountered again later, there is a **secondary immune response**, a response that is faster (typically peaking only 2–7 days after exposure), of greater magnitude, and more prolonged. These differences between primary and secondary immune responses are readily apparent in a graph of the concentrations of specific antibodies in blood over time **(Figure 43.17)**.

Figure 43.17

The specificity of immunological memory.

Primary immune response to antigens A and B produce antibodies slowly with peak concentration of about 10 after about 14 days. Secondary response to antigen A peaks at about 10000 after 7 days.

Long-lived memory cells that are generated in the primary response to antigen A give rise to a heightened secondary response to the same antigen but don’t affect the primary response to another antigen (B).

Graph of primary and secondary immune response in antibody concentration versus time since exposure to antigens A and B. The x-axis is time in days since exposure to antigen A and then to antigens A and B together, ranging from 0 to 56 days with increments every 7 days. The y-axis is antibody concentration (arbitrary units) ranging from 1 to 10000.

- Primary immune response to antigen A produces antibodies to A. Exposure to antigen A first occurs just after Day 0. Production of antibodies to A begins just before Day 7, and peaks after about 14 days at 10 antibody concentration before falling back down to 1 in antibody concentration by Day 28.

- Secondary immune response to antigen A produces antibodies to A; primary immune response to antigen B produces antibodies to B. Exposure to antigens A and B occurs at Day 28. Concentration climbs rapidly for antibodies to A, peaking at nearly 10000 by Day 35, before falling to a concentration of a little above 100 after Day 56. Antibodies to B are not produced until just before day 35, and concentration of antibodies to B peak between Days 41 and 44 at about 10 before falling back to just above 1 after Day 56.

INTERPRET THE DATA *Assume that on average one out of every*105*B cells in the body is specific for antigen A on day*16*and that the number of B cells producing a specific antibody is proportional to the concentration of that antibody. What would you predict is the frequency of B cells specific for antigen A on day*36*?*

The secondary immune response relies on the reservoir of T and B memory cells generated upon initial exposure to an antigen. Because these cells are long-lived, they provide the basis for immunological memory, which can span many decades. (Most effector cells have much shorter life spans, which is why the immune response diminishes after an infection is overcome.) If an antigen is encountered again, memory cells specific for that antigen enable the rapid formation of clones of thousands of effector cells that are also specific for that antigen, thus generating a greatly enhanced immune defense.

Although the processes for antigen recognition, clonal selection, and immunological memory are similar for B cells and T cells, these two classes of lymphocytes fight infection in different ways and in different settings, as we'll explore in Concept 43.3.